

Skills Summary

Tax, Tip, Discount and Markup

Use this reference while completing your worksheet or when studying.

Adding a Percent On: Tax and Tip

Rule: tax and tip are percents *of the price before them*, and they are added on. Two routes give the same number:

- find the extra, then add: $\text{extra} = \frac{p}{100} \times \text{price}$
- multiply once: $\text{total} = \text{price} \times \left(1 + \frac{p}{100}\right)$

Example: A bill is 45 dollars and the tax rate is 8 percent. Find the tax and the total.

Step 1. The tax is a percent of the bill and is added on, so find the extra first and then add it.

Step 2. Tax = $\frac{8}{100} \times 45 = \mathbf{3.6}$ dollars, so the total is $45 + 3.6 = \mathbf{48.6}$ dollars — the same as 45×1.08 .

Try it: A 25 dollar bill gets a 16 percent tip. What is the total paid, in dollars?

check: 67

Taking a Percent Off: Discount and Sale Price

Rule: a discount is a percent *of the original price*, and it is taken off. So d percent off leaves $100 - d$ percent behind: $\text{sale price} = \text{original} \times \left(1 - \frac{d}{100}\right)$. Thirty percent off means you still pay seventy percent.

Example: A coat is marked 80 dollars with 35 percent off. Find the discount and the sale price.

Step 1. The percent is of the marked price, so take 35 percent of 80 and subtract it — or multiply by the 65 percent that is left.

Step 2. Discount = $\frac{35}{100} \times 80 = \mathbf{28}$ dollars, so the sale price is $80 - 28 = \mathbf{52}$ dollars.

Try it: A 90 dollar tent is 20 percent off. What is the sale price, in dollars?

check: 72

Markup, and Working Back to the Original

Rule: a markup is a percent of what the shop *paid*, so selling price = cost $\times (1 + \frac{m}{100})$. Going backwards from the selling price to the cost means **dividing** by that same factor, never subtracting the same percent from the selling price.

Example: A kettle sells for 78 dollars after a 30 percent markup. What did the shop pay, and how large was the markup?

Step 1. The selling price is the cost multiplied by 1.30, so undo the multiplying: divide the selling price by 1.30.

Step 2. Cost = $78 \div 1.30 = 60$ dollars, so the markup itself is $78 - 60 = 18$ dollars.

Try it: A jacket sells for 96 dollars after a 20 percent markup. What did the shop pay, in dollars?

08 :xpep

(!) Watch out: a discount percent is measured against the *original* price, not the sale price, and undoing a markup is a division. Taking 20 percent off 84 dollars gives 67.20, not the 70 dollars the shop actually paid.